Treasury PCA Butterflies, Carry, and Relative-Value Discipline

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Abstract

This paper answers whether a PCA-neutral 2s5s10s Treasury butterfly isolates a tradable curvature residual or only creates a clean-looking factor signal. We estimate Treasury principal components on daily yield changes, construct exact and regularized butterflies, test stationarity and walk-forward hedge stability, and then add carry/rolldown, transaction costs, volatility targeting, purged validation, and actual-bond implementation diagnostics. The evidence is deliberately conservative. The exact full-sample butterfly is neutral to level and slope by construction, but the spread is not strongly stationary, rolling exact hedge ratios become unstable, and stricter mean-reversion variants lose money after realistic filters. The carry-aligned momentum branch is the least-rejected research branch, not a strong strategy result. Its 0.18 annualized Sharpe is too weak for deployment, but useful because it shows which economic direction survives after the naive PCA residual trade fails. That branch earns 307.47 bp in the fixed-maturity research proxy, while CPCV, nested blocked selection, block bootstraps, empirical EGARCH full-curve nulls, and bond-mapping diagnostics keep the claim modest. The trading conclusion is narrower: factor neutrality is not alpha, and promotion to live capital requires executable instruments, financing, costs, and risk controls.

1 Introduction

Treasury relative-value trades often start with a simple desk intuition: broad yield-curve moves can be decomposed into level, slope, and curvature, and a butterfly can isolate what remains. In the 2s5s10s version, the trade owns the wings against the 5Y belly. The construction looks attractive because it appears to remove the obvious macro duration bet while retaining exposure to curve shape.

The problem is that factor neutrality is not the same as edge. A portfolio can have zero exposure to the first two principal components and still fail for reasons that matter in actual rates trading: the residual may not mean-revert, estimated loadings may be sample-dependent, exact hedge ratios may create hidden gross exposure, carry may work against the signal, and the clean fixed-maturity curve may not map cleanly into securities. This paper is built around those failure modes.

The analysis therefore moves in stages. First, it reconstructs the classical PCA butterfly carefully: PCA is run on yield changes rather than yield levels, covariance PCA is used for hedge construction, and exact 2s5s10s weights are solved to neutralize level and slope. Second, it asks whether that residual has usable trading properties. Third, after the naive mean-reversion story weakens, it rebuilds the trade with regularized hedge weights, carry/rolldown, costs, volatility targeting, purged validation, full-curve null simulations, and actual-bond mapping checks.

The so what is direct. A PCA-neutral Treasury butterfly is not alpha by itself. The contribution of this project is showing how a clean curve residual survives or fails once the work imposes carry, hedge instability, turnover, costs, validation discipline, and instrument mapping. In this sample, the naive residual-fading story fails. Regularized hedge construction is cleaner, and carry-aligned curvature momentum is the only branch left worth more execution research. The useful output is not the 0.18 Sharpe. It is the ability to reject the first attractive chart and preserve only the part of the idea that still makes economic sense.

2 Related Literature and Research Design

The factor setup follows Litterman and Scheinkman [1], who show that level, slope, and curvature explain most Treasury yield-curve variation. This paper uses that result as a construction tool rather than a trading conclusion. The return-predictability motivation is related to Cochrane and Piazzesi [2], while the implementation caution is closer to fixed-income arbitrage work such as Duarte, Longstaff, and Yu [3], where apparent relative value must be interpreted through leverage, drawdowns, liquidity, and financing.

The statistical design borrows the discipline of time-ordered validation. Newey and West [4] motivate robust inference when financial labels overlap. Lopez de Prado [5] motivates purging and embargoing folds when strategy labels share future information windows. Engle [6] motivates treating volatility clustering as a central feature of financial returns rather than a nuisance. The GSW fixed-maturity curve [7] provides the clean curve object, while Treasury liquidity work such as Fleming [8] motivates separating fixed-maturity research from actual security execution.

3 Data and Public Research Contract

The data set combines the Gurkaynak–Sack–Wright fixed-maturity Treasury curve with an actual-bond panel used only for implementation diagnostics. These inputs give the two objects needed for the research question: a clean curve for factor construction and a security panel for checking how far the clean curve is from something tradable.

dataset	observations	start	end	columns
GSW fixed-maturity yields	16097	1961-06-14	2025-12-26	30
Treasury bond panel	204868	2022-01-03	2025-12-31	14

Table 1: Data coverage.

The required local workbooks are `gsw_yields.xlsx` and `treasury_panel_pca.xlsx`. Raw workbooks stay outside version control. The pipeline validates their schema and regenerates the tables, figures, integrity checks, and artifact manifest.

4 PCA and Butterfly Construction

Let Δy_t denote the vector of daily Treasury yield changes in basis points. The baseline PCA is

$$\Delta y_t = L f_t + \epsilon_t, \tag{1}$$

where the columns of L are factor loadings and f_t are daily curve shocks. The paper compares correlation PCA for interpretation with covariance PCA for hedge construction. Covariance PCA keeps yield-basis-point units intact, which is the natural unit for spread construction and P&L

proxies.

factor	correlation_pca_variance_pct	covariance_pca_variance_pct
PC1	88.0866	88.1190
PC2	8.6741	8.6887
PC3	2.3239	2.2753

Table 2: Explained variance from daily yield-change PCA.

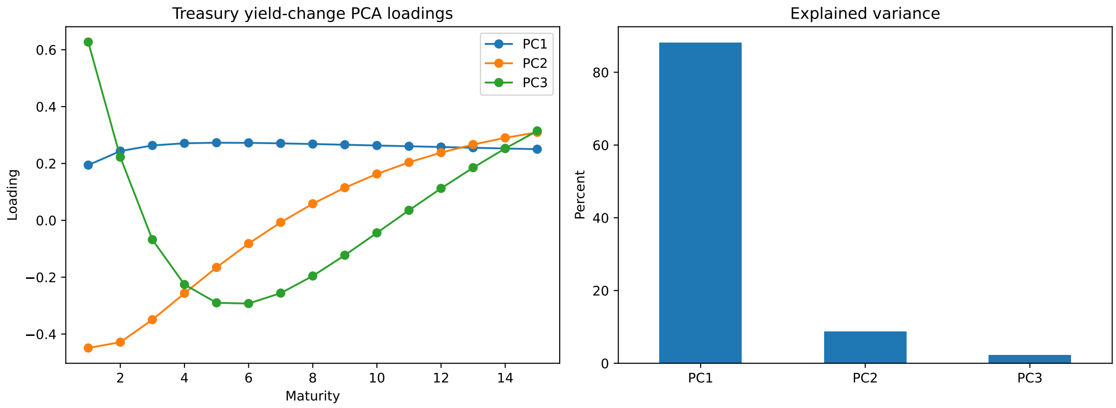

Figure 1: Covariance PCA loadings and explained variance. The first three components recover the standard level, slope, and curvature shape.

The exact 2s5s10s butterfly solves for weights $w = (w_2, w_5, w_{10})$ such that

$$w^\top L_1 = 0, \qquad w^\top L_2 = 0, \qquad w_5 = -1. \tag{2}$$

The spread is

$$s_t = w_2 y_{2,t} + w_5 y_{5,t} + w_{10} y_{10,t}. \tag{3}$$

The exact weights and regularized weights are reported together because their difference is the central implementation lesson.

maturity	exact_weight	regularized_weight
2.0000	0.5777	0.4047
5.0000	-1.0000	-1.0000
10.0000	0.5026	0.2407

Table 3: Exact PC-neutral weights versus regularized weights. The belly is normalized to -1.

factor	exact_exposure	regularized_exposure
PC1	0.0000	-0.1111
PC2	0.0000	0.0314
PC3	0.3963	0.3694

Table 4: PC exposures of exact and regularized butterfly weights. Regularization accepts small level/slope exposure in exchange for lower gross and more stable weights.

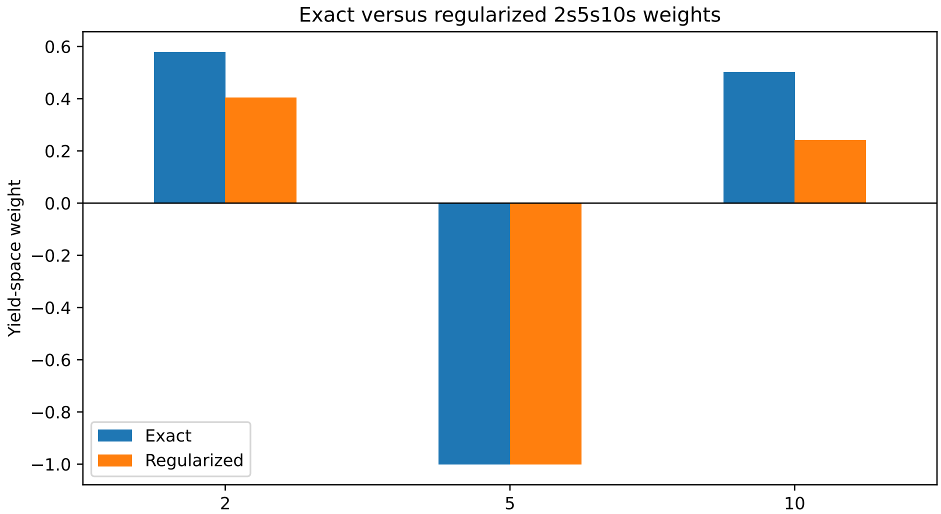

Figure 2: Exact PC-neutral weights versus regularized weights.

5 Stationarity, Hedge Stability, and Carry

The exact construction works mechanically: it neutralizes PC1 and PC2 and keeps curvature exposure. That is not enough. The exact static spread has an ADF p-value of 0.1669 and an estimated half-life of about 1,222 trading days. The regularized spread is even more persistent by this diagnostic. A residual this slow should not be treated as a high-confidence mean-reversion trade.

spread	adf_stat	pvalue	used_lag	nobs	half_life_days
exact_static	-2.3160	0.1669	0.0000	16096.0000	1222.0722
regularized_static	-1.6451	0.4596	35.0000	16061.0000	1773.6346

Table 5: Stationarity diagnostics for exact and regularized fixed-maturity spreads.

Walk-forward estimation exposes the second problem. Exact rolling PCA weights can become very large because local loadings shift. The full-sample exact trade is elegant; the rolling exact trade is fragile. Regularization changes the decision from "zero exposure at any cost" to "small residual exposure with bounded gross." The monthly regularized hedge averages 0.3993, -1.0000, and 0.2363 on the 2Y, 5Y, and 10Y legs, with average gross exposure of 1.6356 and a maximum of 1.8607.

metric	count	mean	std	min	25%	50%	75%	max
2	623.0000	0.3993	0.0859	0.1811	0.3484	0.4216	0.4660	0.5373
5	623.0000	-1.0000	0.0000	-1.0000	-1.0000	-1.0000	-1.0000	-1.0000
10	623.0000	0.2363	0.0664	0.0467	0.2069	0.2333	0.2659	0.4279
PC1_exposure	623.0000	-0.0873	0.0206	-0.1305	-0.1008	-0.0891	-0.0731	-0.0329
PC2_exposure	623.0000	0.0718	0.0602	-0.0731	0.0440	0.0698	0.1107	0.2163
gross	623.0000	1.6356	0.1110	1.3751	1.5868	1.6332	1.7063	1.8607

Table 6: Walk-forward regularized weight and residual exposure summary.

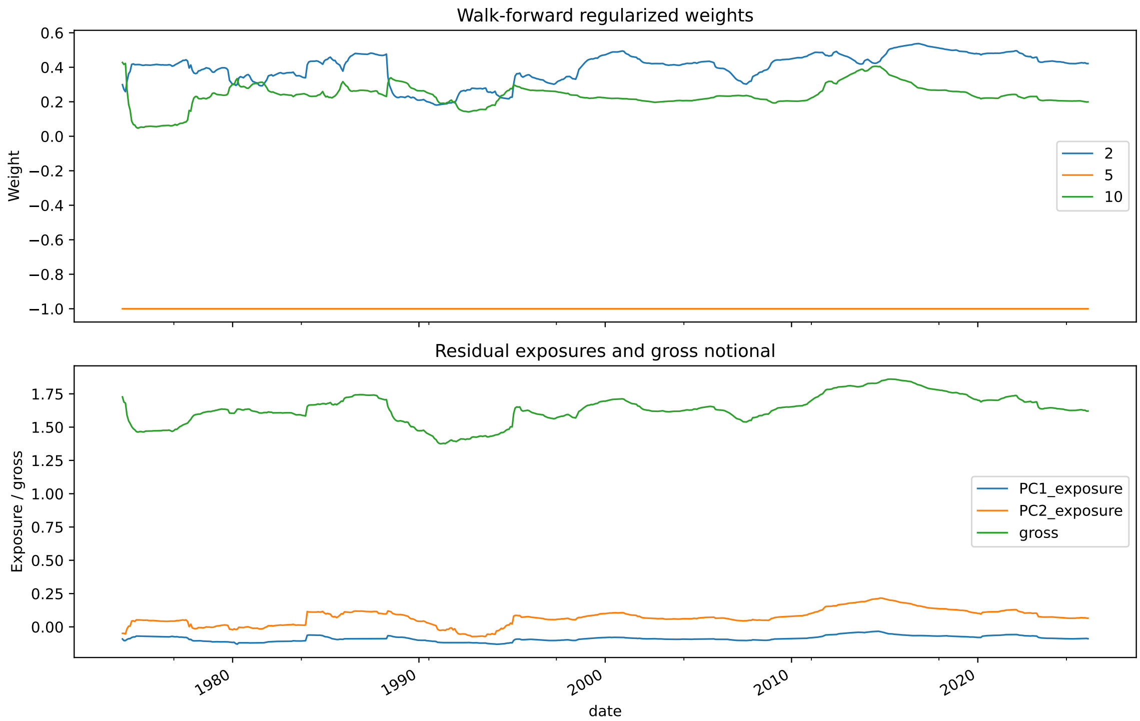

Figure 3: Walk-forward regularized weights, residual PC exposures, and gross notional.

Carry enters because a curve trade can look statistically cheap while the curve's own rolldown pushes against it. The research proxy uses unchanged-curve rolldown of the weighted yield spread. It is not full bond carry, but it is enough to prevent the strategy from blindly fading a persistent spread when the curve shape pays the opposite direction.

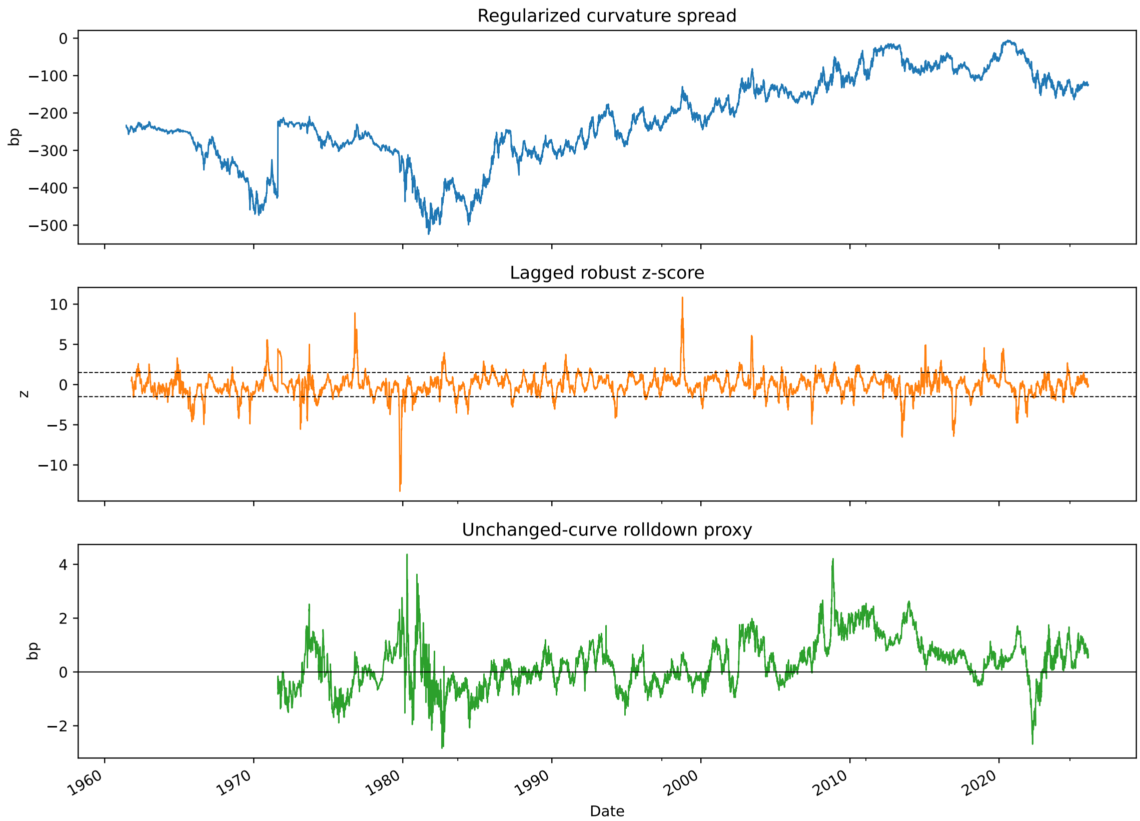

Figure 4: Regularized curvature spread, lagged robust z-score, and unchanged-curve rolldown proxy.

6 Investability Screen

The strategy layer tests two hypotheses. The first is residual mean reversion: fade rich or cheap curvature after robust z-score, carry gating, volatility targeting, maximum holding periods, and transaction costs. The second is carry-aligned curvature momentum: if the spread has been moving and rolldown points the same way, follow the move rather than fight it.

strategy	total_pnl_bp	ann_sharpe	max_drawdown_bp	hit_rate_active	active_days	trades
Exact static mean reversion	303.3543	0.2005	-158.3710	0.5033	10736.0000	617.0000
Exact rolling mean reversion	988.6256	0.0227	-3851.7434	0.4812	9054.0000	440.0000
Regularized mean reversion	-372.6911	-0.3577	-454.1899	0.4925	2057.0000	1705.0000
Rolling regularized mean reversion	-449.2806	-0.4985	-493.2492	0.4965	2117.0000	1916.0000
63-day carry-aligned momentum	307.4696	0.1837	-125.1415	0.5108	7978.0000	7345.0000

Table 7: Strategy diagnostics. P&L is a fixed-maturity research proxy in yield-spread bp.

The table changes the story. Exact static mean reversion earns 303.35 bp with a 0.20 Sharpe, but that result uses full-sample weights. Exact rolling mean reversion earns 988.63 bp with a near-zero 0.02 Sharpe and a -3,851.74 bp drawdown, so total P&L is the wrong headline. The stricter regularized mean-reversion variants lose money after costs. The carry-aligned momentum branch is the least-rejected branch, not a strong strategy result. It earns 307.47 bp with a 0.18 Sharpe and -125.14 bp max drawdown.

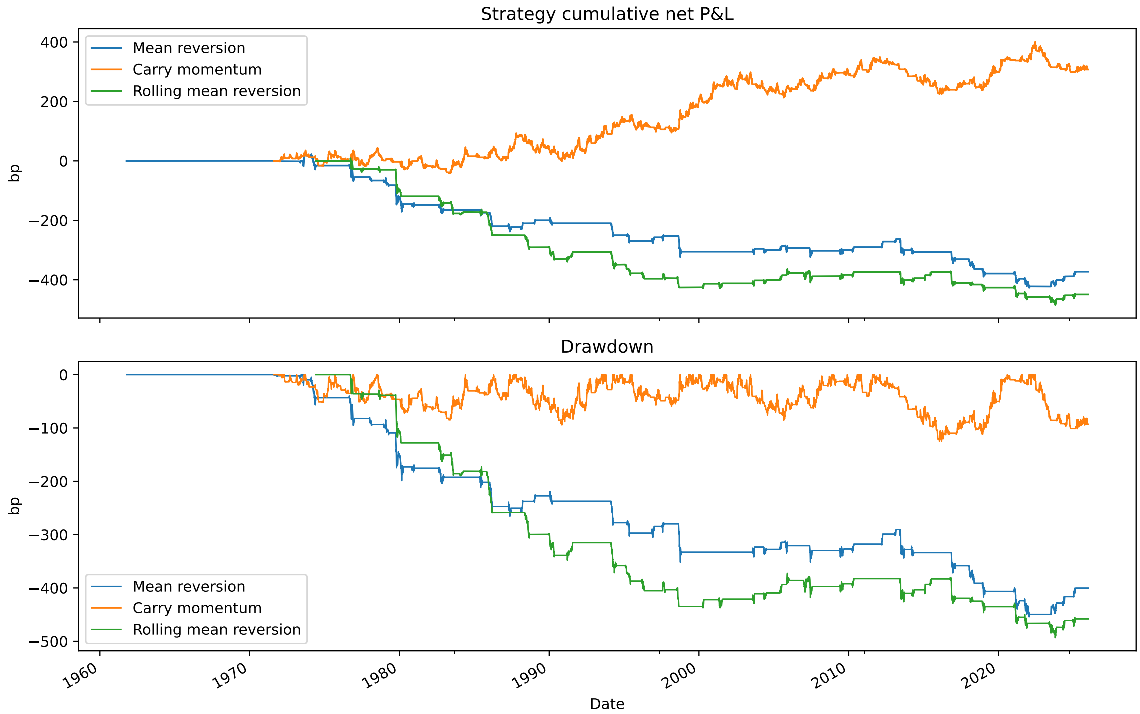

Figure 5: Cumulative net P&L and drawdown for regularized strategy hypotheses.

These are not live-trading statistics. They are fixed-maturity, yield-space research diagnostics. Their purpose is to decide which hypothesis deserves instrument-level work. On that standard, mean reversion is rejected and carry-aligned momentum remains alive but modest.

7 Robustness and Anti-Overfit Controls

The main overfit risk is strategy selection. We therefore evaluate pre-declared mean-reversion configurations and momentum horizons with chronological combinatorial purged cross-validation. The purge length is 42 trading days, matching the maximum holding period, with an additional five-day embargo around test windows. The folds are interpreted as Treasury-curve regime blocks, not IID samples.

strategy_family	config_id	lookback	entry_z	transaction_cost_bp	folds	mean_test_n	mean_total_pnl_bp	median_total_pnl_bp	positive_fold_rate	mean_ann_sharpe	worst_drawdown_bp	mean_trades
carry_aligned_momentum	21.0000	21.0000	NaN	0.0300	28.0000	3390.2500	81.3362	103.4523	0.7500	0.1943	-175.2698	1797.5000
carry_aligned_momentum	63.0000	63.0000	NaN	0.0300	28.0000	3390.2500	76.8674	86.3671	0.8571	0.1776	-182.9832	1836.2500
carry_aligned_momentum	126.0000	126.0000	NaN	0.0300	28.0000	3390.2500	-3.3284	-1.9892	0.4643	-0.0060	-199.9726	1832.2500
carry_gated_mean_reversion	24.0000	252.0000	2.0000	0.0100	28.0000	3390.2500	-24.2439	-25.3442	0.3571	-0.0592	-179.4367	191.5000
carry_gated_mean_reversion	25.0000	252.0000	2.0000	0.0300	28.0000	3390.2500	-24.5261	-25.7219	0.2857	-0.0609	-179.8095	191.5000
carry_gated_mean_reversion	26.0000	252.0000	2.0000	0.0500	28.0000	3390.2500	-24.8083	-26.0997	0.2857	-0.0625	-180.1823	191.5000
carry_gated_mean_reversion	21.0000	252.0000	1.5000	0.0100	28.0000	3390.2500	-40.6162	-50.2361	0.3214	-0.1637	-181.2359	324.0000
carry_gated_mean_reversion	22.0000	252.0000	1.5000	0.0300	28.0000	3390.2500	-41.0694	-50.8306	0.3214	-0.1659	-181.6708	324.0000
carry_gated_mean_reversion	23.0000	252.0000	1.5000	0.0500	28.0000	3390.2500	-41.5225	-51.4251	0.3214	-0.1681	-182.1056	324.0000
carry_gated_mean_reversion	18.0000	252.0000	1.0000	0.0100	28.0000	3390.2500	-51.7276	-50.4308	0.2500	-0.1970	-219.3557	554.7500
carry_gated_mean_reversion	19.0000	252.0000	1.0000	0.0300	28.0000	3390.2500	-52.5012	-51.2239	0.2500	-0.2001	-219.9669	554.7500
carry_gated_mean_reversion	20.0000	252.0000	1.0000	0.0500	28.0000	3390.2500	-53.2749	-52.0170	0.2500	-0.2033	-220.5781	554.7500

Table 8: CPCV strategy diagnostics across pre-declared strategy families.

The CPCV table reinforces the economic conclusion. Carry-aligned momentum ranks above the mean-reversion family. The 21-day momentum variant has mean fold P&L of 81.34 bp and positive P&L in 75% of folds; the 63-day variant has mean fold P&L of 76.87 bp and positive P&L in 85.71% of folds. The best mean-reversion families remain negative.

null	observed_total_pnl_bp	observed_ann_sharpe	pvalue_total_pnl	pvalue_ann_sharpe	mean_sim_total_pnl_bp	median_sim_total_pnl_bp
factor_residual_block_bootstrap	307.4696	0.1837	0.0800	0.0800	49.9459	65.7351
empirical_egarch_full_curve	274.3035	0.2128	0.0900	0.1000	15.1845	2.7758

Table 9: Null simulations for the carry-aligned momentum branch. The full-curve null uses empirical heavy-tailed innovations and EGARCH-style volatility clustering.

series	student_t_df_mle
2Y yield changes	3.2826
5Y yield changes	4.4958
10Y yield changes	5.2908

Table 10: Student-t tail diagnostics for daily yield changes.

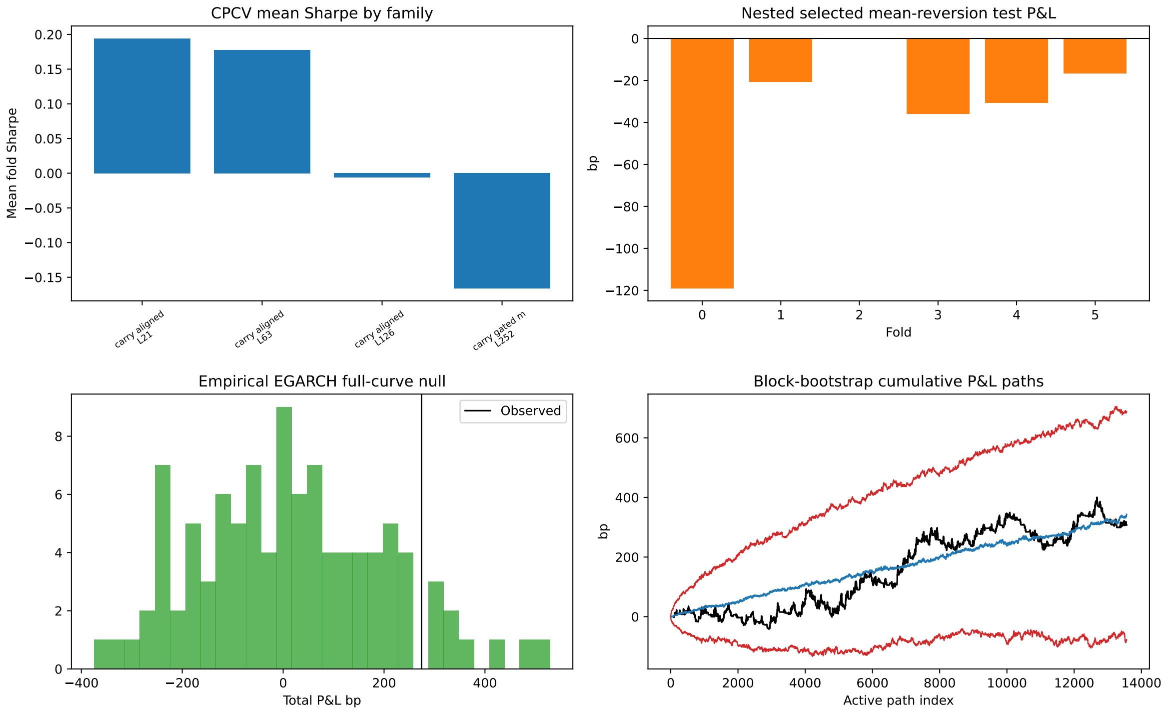

Figure 6: Robustness dashboard: CPCV strategy ranking, nested selected mean-reversion tests, empirical EGARCH full-curve nulls, and block-bootstrap cumulative P&L paths.

The validation layer is intentionally conservative. Under the factor-residual block bootstrap, the observed carry-momentum P&L has one-sided p-values of 0.08 on total P&L and 0.08 on Sharpe. Under the empirical EGARCH full-curve null, the p-values are 0.09 and 0.10 in the regenerated public artifacts. This is encouraging enough to keep the branch alive, but not strong enough to call it validated production alpha.

8 Bond Implementation and Production Gap

Fixed-maturity yields are clean research objects. Trades are done in bonds, futures, swaps, or packages. The repo therefore maps monthly rebalances to nearby Treasury bonds around the 2Y, 5Y, and 10Y targets and compares selected-bond yield changes against GSW fixed-maturity changes. This is where many curve-RV projects become unrealistic.

target_tenor	n	mean_abs_ttm_error	max_abs_ttm_error	mean_duration	mean_dirty_price
2.0000	48.0000	0.0043	0.0096	1.9684	96.2683
5.0000	48.0000	0.0043	0.0089	4.7616	92.9824
10.0000	48.0000	0.1292	0.2204	8.3385	99.5173

Table 11: Nearest-bond mapping quality by target tenor.

target_tenor	n	mean_tracking_error_bp	median_tracking_error_bp	rmse_tracking_error_bp	p95_abs_tracking_error_bp
2.0000	48.0000	-0.2188	0.5148	8.4073	17.3374
5.0000	48.0000	-0.1394	0.0831	8.0000	18.0569
10.0000	48.0000	-0.2179	0.2127	7.3476	14.9740

Table 12: Selected-bond yield-change tracking error versus fixed-maturity targets.

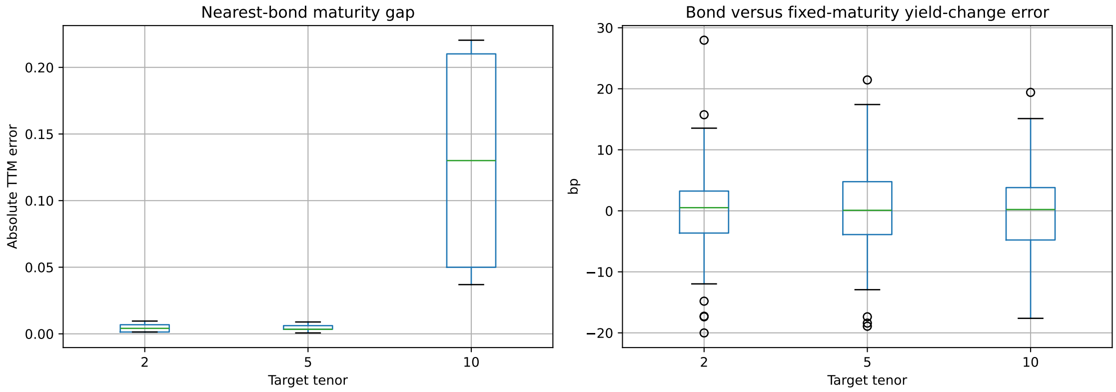

Figure 7: Actual-bond implementation diagnostics: maturity gap and fixed-maturity tracking error.

The tracking error is not catastrophic, but it is not zero. The 2Y, 5Y, and 10Y selected-bond RMSEs are about 8.41, 8.00, and 7.35 bp. A production version would need DV01 sizing, coupon and benchmark selection, bid/ask, repo and financing, accrued interest, roll mechanics, futures CTD logic if implemented through futures, and portfolio-level risk controls. Those are not cosmetic details. They decide whether the fixed-maturity signal survives the instrument layer.

9 Conclusion

Through this sequence, we can conclude that a PCA-neutral Treasury butterfly is a useful construction but not a strategy by itself. The factor model works: Treasury yield changes are dominated by level, slope, and curvature, and exact 2s5s10s weights can neutralize level and slope. The trade thesis is weaker. The residual is not strongly stationary, exact rolling hedge ratios are unstable, and stricter mean-reversion variants lose money after costs.

The final conclusion is bounded. Regularized hedge construction produces stable weights, and carry-aligned curvature momentum is more plausible than naive z-score fading. The anti-overfit layer keeps that branch alive but does not promote it to live alpha. A production version would need DV01 sizing, benchmark or CTD selection, bid/ask, repo and financing, roll mechanics, and portfolio-level risk controls. The result is useful because it shows the full research path: build the factor trade, test the attractive story, reject what fails, and state clearly what still has to be true before capital can be put behind it.

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